HEB-ACC-43 Arista Access Node QoS Congestion Event Warning Alarm (ALARM_1072) Resolution SOP

Node: HEB-ACC-43 | **Vendor:** Arista | **Severity:** Warning | **Alarm Code:** ALARM_1072

Description

This Standard Operating Procedure outlines the necessary steps to diagnose and resolve a QoS Congestion Event Warning Alarm (ALARM_1072) on the HEB-ACC-43 Arista Access Node. The failure domain in question revolves around microbursts, interface tail drops, and QoS queue starvation. In Arista equipment, access nodes are responsible for providing high-speed connectivity to edge devices. As such, they are susceptible to congestion events, particularly when dealing with high-bandwidth applications. The underlying protocol mechanics that contribute to this failure domain include the prioritization of packets through the use of Differentiated Services Code Points (DSCPs) and the scheduling of packets through the use of Weighted Random Early Detection (WRED). Furthermore, the hardware mechanics of the Arista access node, including the line card and fabric interfaces, play a crucial role in determining the node's ability to handle congestion events. A combination of these factors can lead to microbursts, interface tail drops, and QoS queue starvation, ultimately resulting in the ALARM_1072 warning alarm.

Pre-requisite Checks

- Verify that the Hebbal Edge site is accessible and that the HEB-ACC-43 node is powered on.

- Check the status of the redundant link to ensure that it is operational and not contributing to the congestion event.

- Review the network topology to identify any potential bottlenecks or misconfigured QoS policies that may be contributing to the congestion event.

Potential Root Causes

Cause	Probability
Misconfigured QoS policy leading to queue starvation	High
Insufficient buffer space on the line card leading to microbursts	Medium
Configuration error on the fabric interface leading to interface tail drops	Low

Diagnostic Commands & Expected Outputs

```
Command: show qos policy summary, Expected Output: Policy name
Priority Bandwidth DSCP, Description: This command provides an
overview of the QoS policies configured on the node. It helps to
identify any misconfigured policies that may be contributing to the
congestion event.
```

```
Command: show interface statistics, Expected Output: Interface Bytes Packets Errors Drops, Description: This command provides detailed statistics on the interface, including the number of packets dropped due to congestion. It helps to identify if the interface is experiencing tail drops.

Command: show buffer utilization, Expected Output: Buffer Used Total, Description: This command provides information on the buffer utilization on the line card. It helps to identify if there is sufficient buffer space to handle the incoming traffic.
```

Resolution Steps

Step 1: Step: Verify QoS policy configuration, Command: show qos policy detail, Description: Review the QoS policy configuration to ensure that it is correctly configured and not contributing to the congestion event.

Step 2: Step: Adjust QoS policy configuration, Command: config qos policy {policy-name}, Description: Adjust the QoS policy configuration to ensure that it is correctly prioritizing packets and preventing queue starvation.

Step 3: Step: Verify buffer space allocation, Command: show buffer allocation, Description: Verify that sufficient buffer space is allocated to handle the incoming traffic.

Step 4: Step: Adjust buffer space allocation, Command: config buffer allocation {buffer-name} {size}, Description: Adjust the buffer space allocation to ensure that it is sufficient to handle the incoming traffic.

Post-Check Verification

```
Command: show qos policy summary, Expected Output: Policy name Priority Bandwidth DSCP, Description: Verify that the QoS policy configuration is correctly configured and not contributing to the congestion event.

Command: show interface statistics, Expected Output: Interface Bytes Packets Errors Drops, Description: Verify that the interface is not experiencing tail drops.

Command: show buffer utilization, Expected Output: Buffer Used Total, Description: Verify that sufficient buffer space is allocated to handle the incoming traffic.
```

Roll-back Procedures

- Step: Revert QoS policy configuration, Command: rollback qos policy {policy-name}, Description: Revert the QoS policy configuration to its previous state in case the fix fails.

- Step: Revert buffer space allocation, Command: rollback buffer allocation {buffer-name}, Description: Revert the buffer space allocation to its previous state in case the fix fails.